

Bosch Ltd.

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Project Trainee/ 2020BME065

Nashik Plant (NaP) | JAN - MAY

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Fixture Design for A/F milling ANGLE BAD



BOSCH

NaP/ MFC 2

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Bosch in India



1922

1st Sales Office
in Calcutta

1953

1st Manufacturing
Operations
in Bangalore

Bosch Ltd

15
Group Companies

15

16

Manufacturing
Facilities

6 manufacturing
Facilities of Bosch Ltd

195

Billion rupees
Sales Revenue
In 2023

Bosch Ltd **100**
Billion rupees

INJECTOR MANUFACTURING PLANT NaP

3,500

Sales Outlet

1,500

Suppliers

Associates

31,530

Bosch Ltd Associates

7,550

MOBILITY
SOLUTIONS

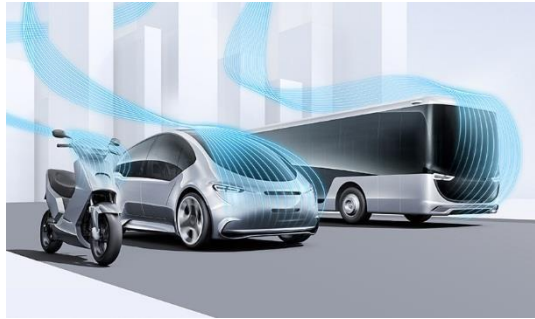
ENERGY &
BUILDING TECH.

CONSUMER
GOODS

INDUST RIAL
TECH

RBE&BS - Integrated system solutions

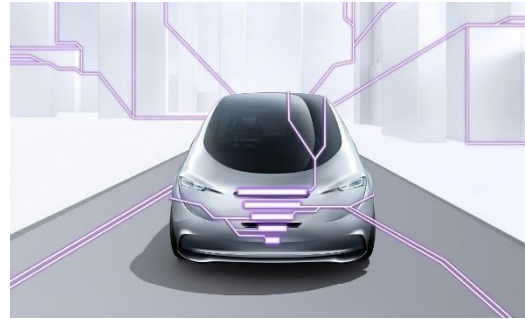
Personalized mobility



Automated mobility



Connected mobility



Powertrain systems



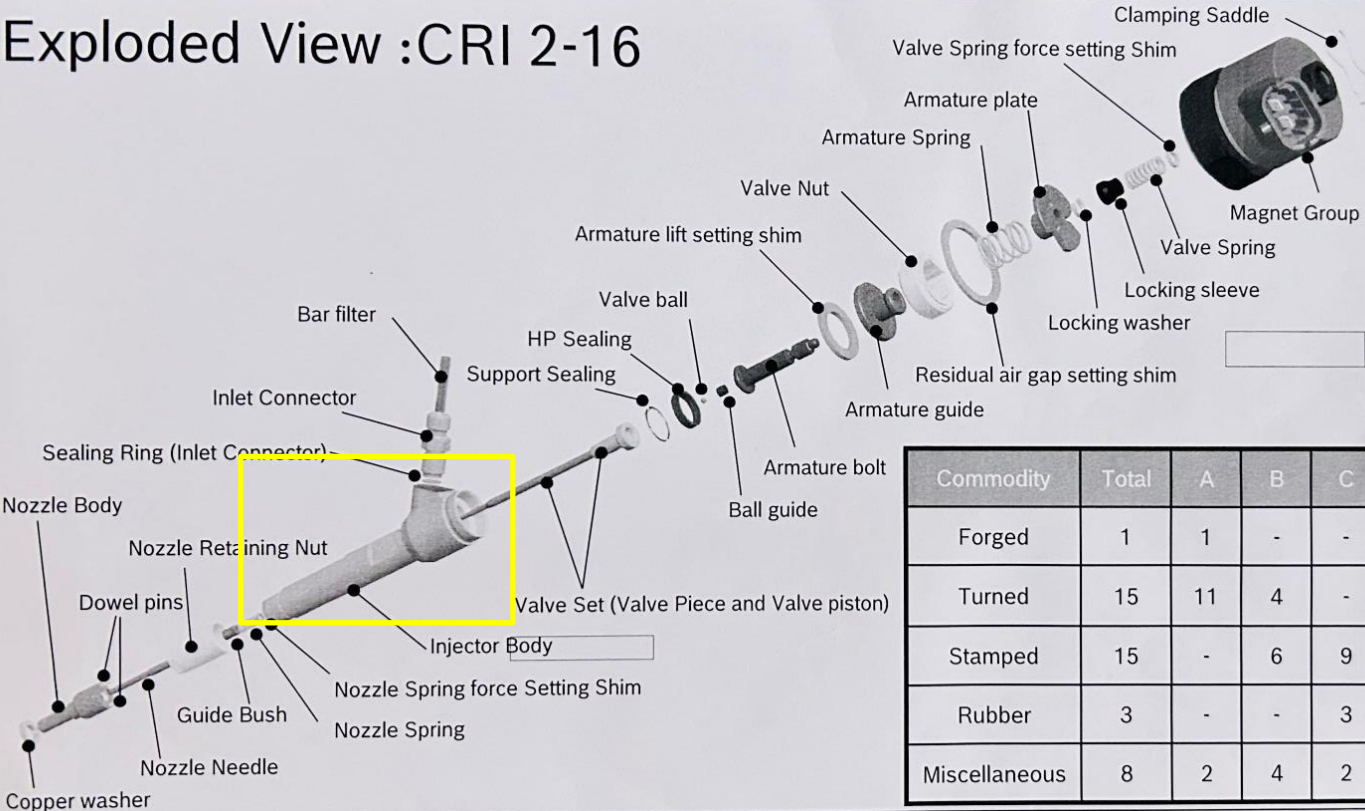
**fun and
fascinating**

**safe and
comfortable**

**efficient and
economical**

COMMON RAIL INJECTOR (CRI)

Exploded View :CRI 2-16



Commodity	Total	A	B	C
Forged	1	1	-	-
Turned	15	11	4	-
Stamped	15	-	6	9
Rubber	3	-	-	3
Miscellaneous	8	2	4	2

Departments – Bosch (NaP)

Bosch Limited

MFC

HR

QMM

HS&E

TEF

MFN

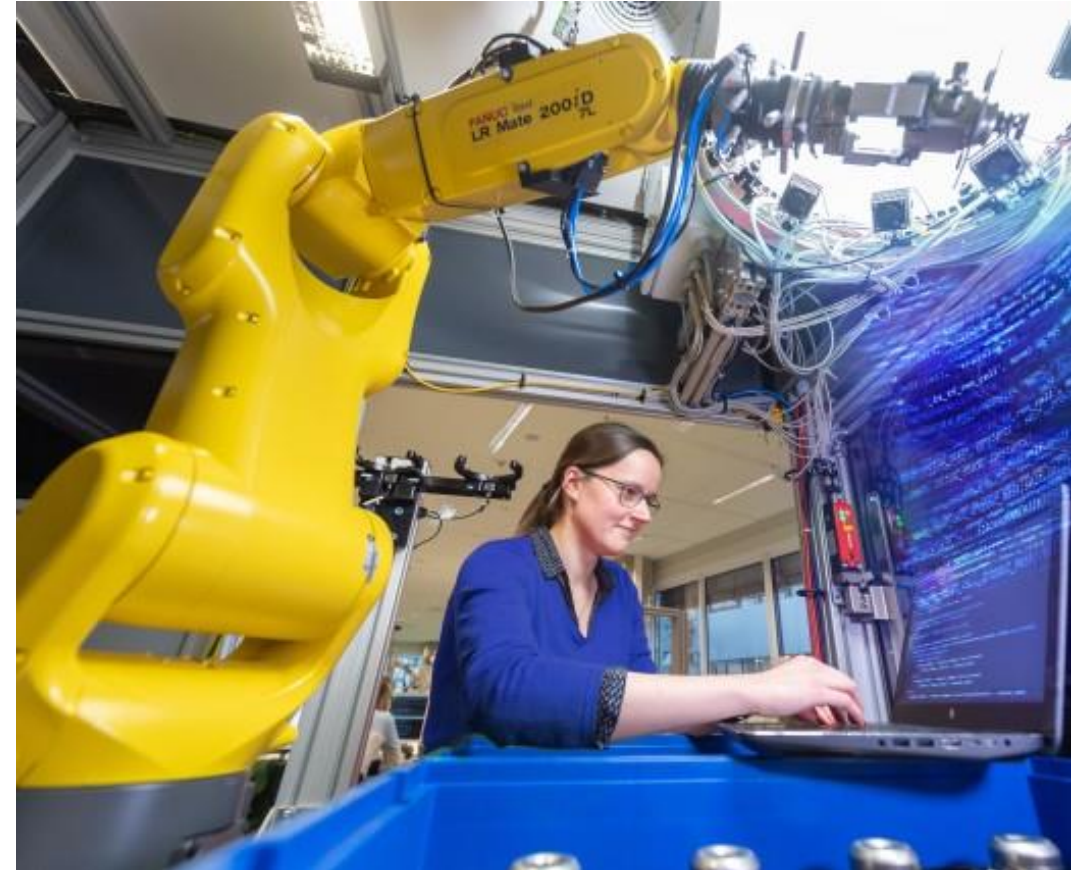
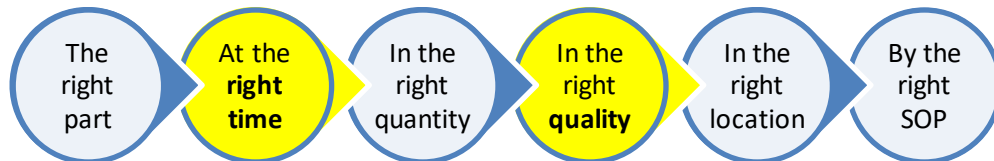


MFC - DEPT

MFC stands for **MANUFACTURING OF COMMON RAIL INJECTOR**, is the dept at Bosch Nashik plant which deals with the production of various common rail injectors.



Objectives of MFC: To Deliver,



PFD – Milling station

FIFO (I)

Milling m/c

Inspection by
Gauging

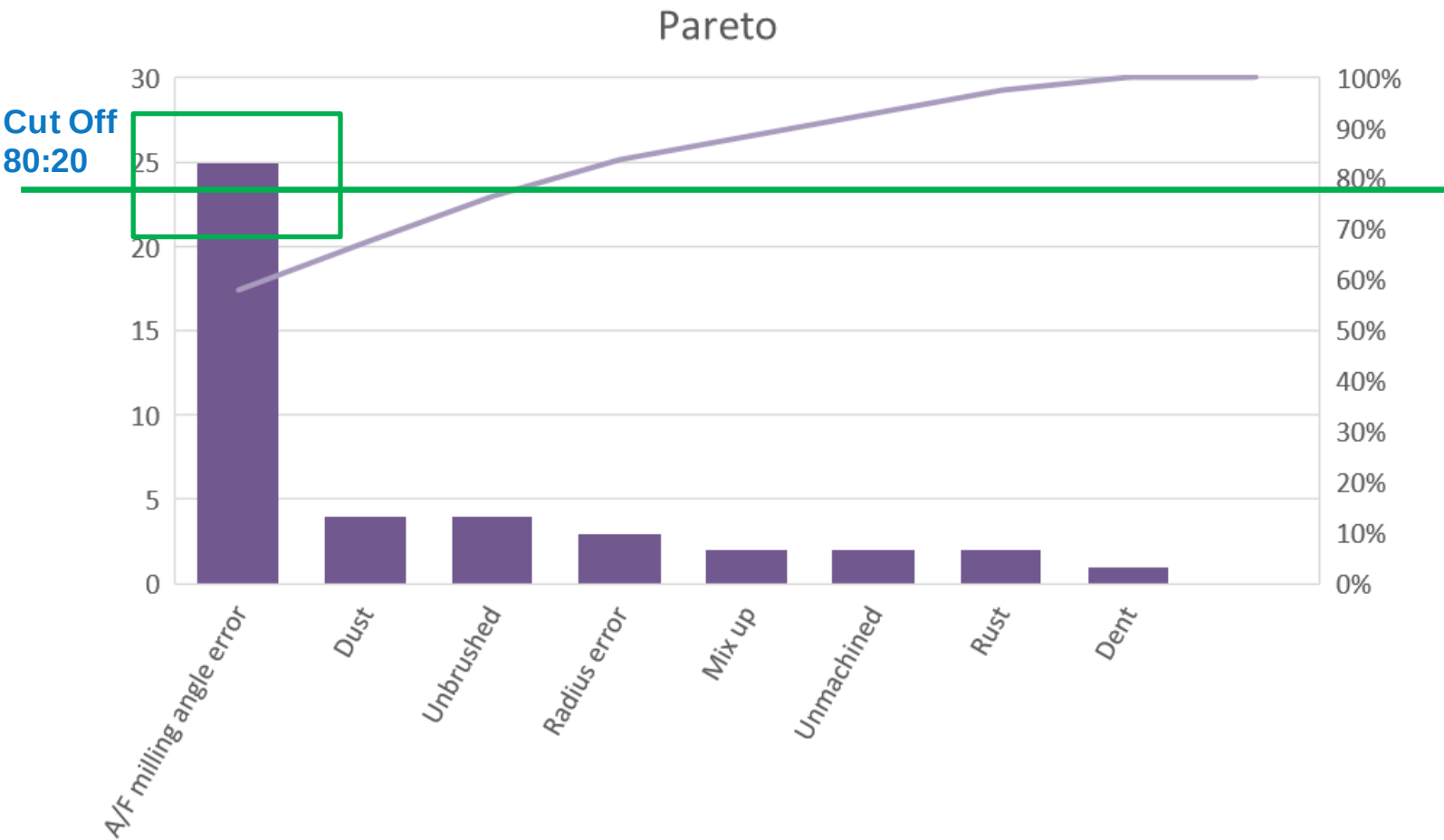
Brushing
process

Declamping process

FIFO (O)



Complaints – Assembly line of Injector



Problem statement

Across Flat Milling Angle Errors:

Significant deviations in milling angles during the across flat milling operation, surpassing acceptable tolerances.

Technical Constraints:

1. Structural Complexity:

Injector body design is standardized globally, restricting modifications for injector and Gauging parameters too.

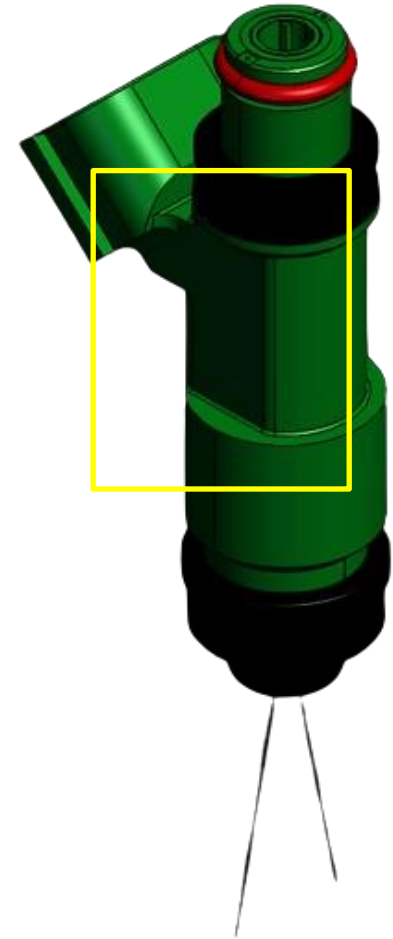
2. High-Pressure Environment:

Inability to place detection fixtures within the milling machine due to operational pressures.

Approach:

Development of a detection fixture for the declamping process to be installed on the declamping machine, utilizing robust materials and precise engineering to detect milling angle errors without additional damage to the injector bodies.

CRI – Isometric view

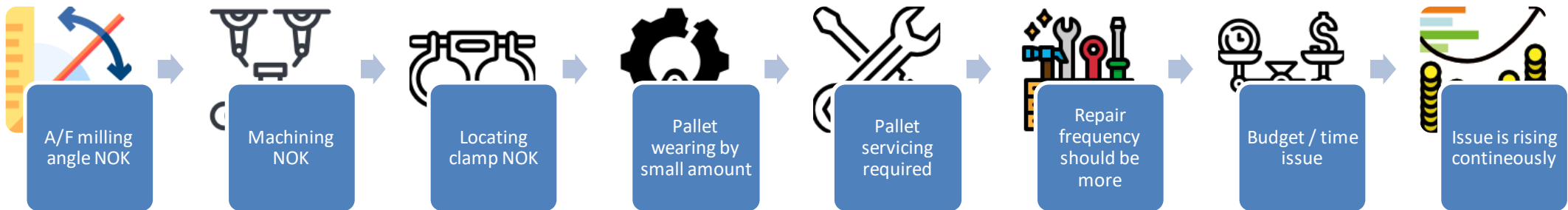


Root Cause Analysis – A/F milling angle NOK

To find the root cause analysis, let's implement POKA YOKE (Mistake proofing) for existing system. There are two types of POKA YOKE that should be implemented **chronologically**.

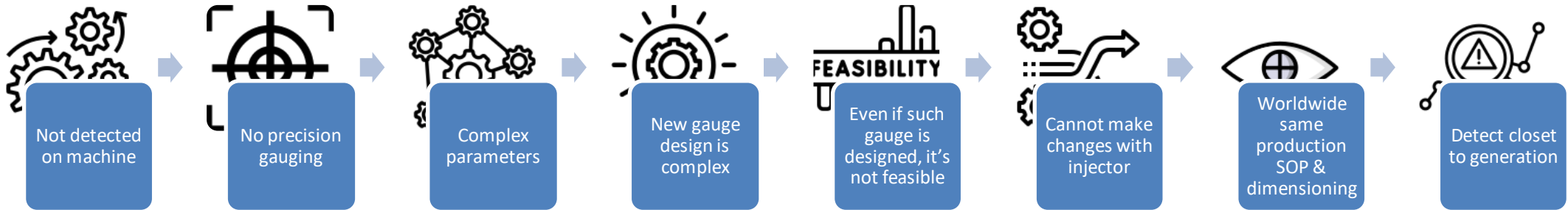
- 1. GENERATION poka yoke - to avoid generating the defect.
- 2. DETECTION poka yoke - to detect if generated.

Generation POKA YOKE



Detection POKA YOKE :

Even if the defect is generated it should be detected where it is generated.



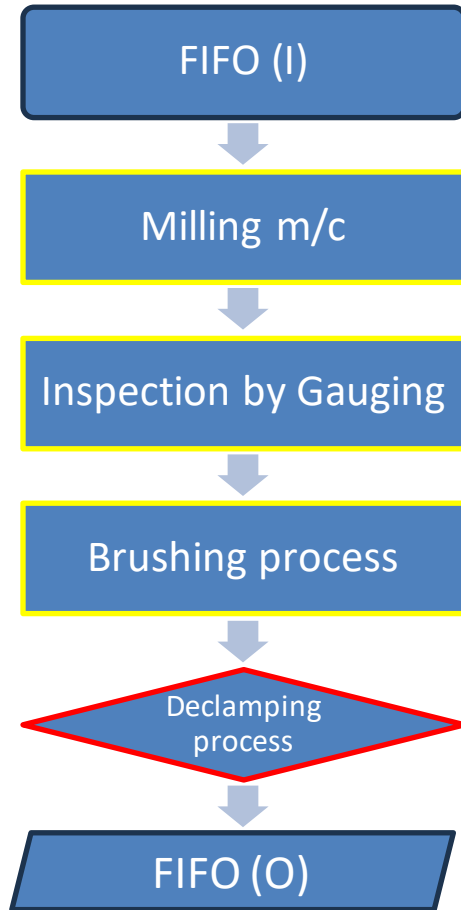
Objective:

To detect and isolate defective injector bodies with across flat milling angle errors during the declamping process, preventing them from reaching the assembly line.

Rationale:

Identifying defects closest to their point of generation to minimize impact on downstream processes.

PROCESS FLOW DIAGRAM – MILLING STATION



Detection stages:

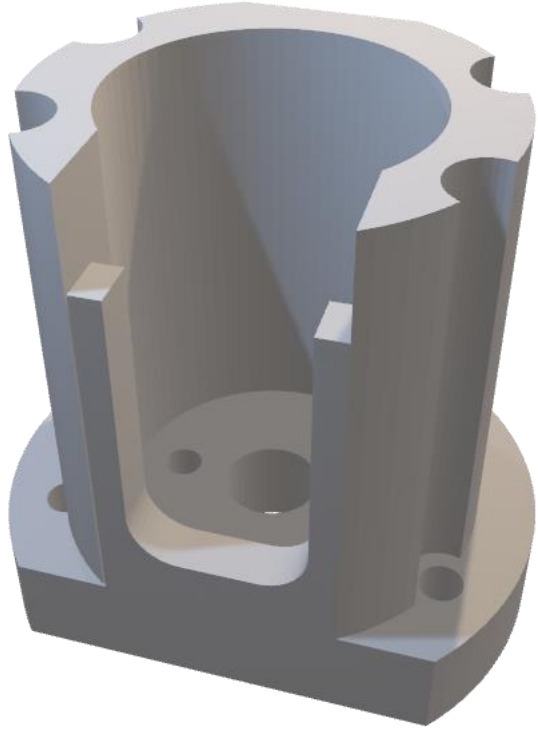
1. At FIFO input: Cannot detect as process not started. ✗
2. At Milling machine: Cannot detect inside machine because machine geometry and high pressure inside m/c. ✗
3. At Gauging: Cannot detect due to gauge inaccuracy. ✗
4. At brushing: Enclosed Robotic arm. ✗
5. At declamping: Can be detected as pressure controllable and machine is handy. ✓

Solution Approach

Fixture Design: Development of a detection fixture for the declamping process, utilizing robust materials and precise engineering to detect milling angle errors without additional damage to the injector bodies.

FIXTURE DESIGN

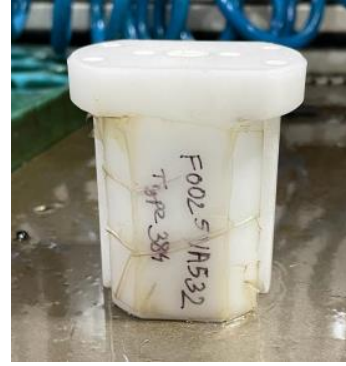
Creating a fixture with respect to the dimensions and tolerances of injector body



3-D View



Front view



Rear view



Top View

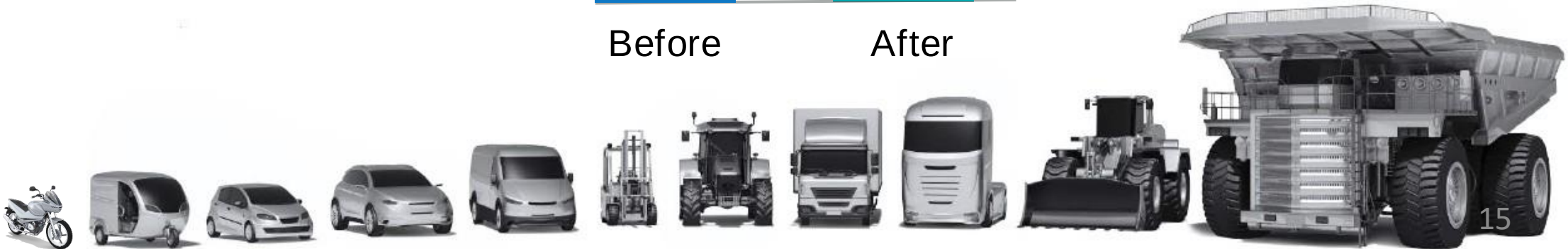
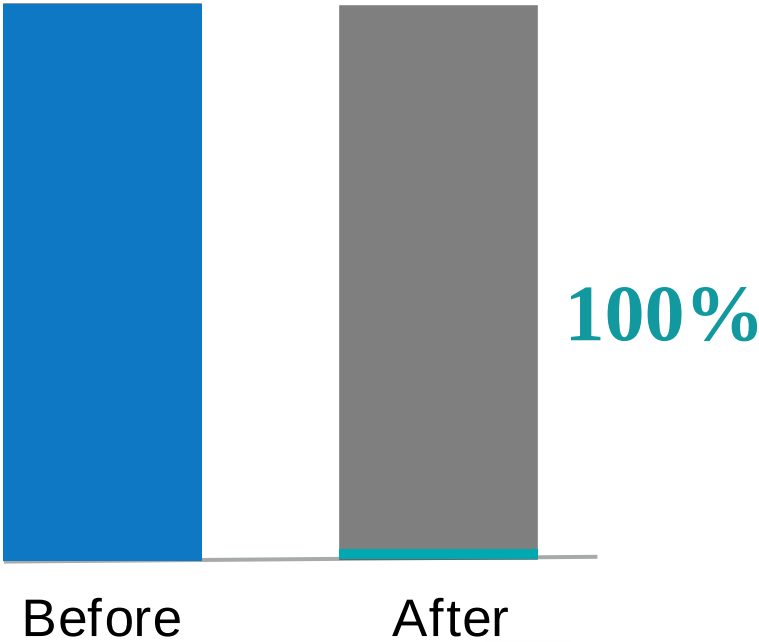


Bottom view



Results

A/F milling issues detection - Errors



Future Scope

If time and money invested,

Automated servicing station

- Development of automated stations for regular maintenance of clamping pallets to prevent wear and damage.

Smart clamping pallets

- pallets with embedded sensors and IoT capabilities to monitor wear and alert for maintenance proactively.

Advanced sensors

- Integration of high-precision optical and laser sensors for enhanced accuracy in detecting milling defects.

Adaptive fixtures

- Development of adaptive fixtures that can adjust to different injector body types and configurations automatically.

Integrated QC

- Real-time integration of quality control systems within the milling process for immediate error detection and correction.





Thank-you!!